গবেষণা, পরীক্ষণ ও কনসালটেনসি কেন্দ্র
পুর ও পরিবেশ কৌশল বিভাগ
শাহজালাল বিজ্ঞান ও প্রযুক্তি বিশ্ববিদ্যালয়
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# SUB-SOIL INVESTIGATION REPORT

**FOR PROPOSED 05 STORIED RESIDENTIAL BUILDING CONSTRUCTION AT VILL-LIAKOTPUR, BIROSREE, ZOKIGONJ, SYLHET. DAG # 325, KHA # 6694, J.L # 70, MOUJA- BOHOR.**

## CLIENT

**MD. ZAMIL AHMED CHOWDHURY**
**S/O- ALHAJ MD. AZIZUR RAHMAN CHOWDHURY**

Verification Code: C3um91Fc
Verify report at: crtc-cee-sust.com

## CONDUCTED BY

CENTER FOR RESEARCH TESTING & CONSULTANCY
DEPARTMENT OF CIVIL & ENVIRONMENTAL ENGINEERING
SHAHJALAL UNIVERSITY OF SCIENCE & TECHNOLOGY, SYLHET

## SAMPLE COLLECTED BY

**M/S TECHNO SOIL INVESTIGATOR**
Mokothchhor Market, Bagbari Barnomala Point, Sylhet.
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## Introduction

A reasonably accurate conception about the sub-soil parameters of any project site is an essential prerequisite for proper planning and designing the foundation of the structure so that the same after its construction would remain safe and serviceable. A detailed study on the sub-surface elements in this regard would necessary to select proper type of foundation.

## Objectives

The purpose of sub-soil investigation program is to obtain the following information

1. To classify the formation of soil layers
2. To estimate the safe bearing capacities of the foundation at different soil layers
3. To suggest the safe and appropriate type of foundations.

## Scope of work

Centre for research, testing and consultancy (CRTC), Department of Civil and Environmental Engineering, Shahjalal University of Science & Technology (SUST), Sylhet; execute only the laboratory tests and other computational works with the supplied disturbed and/or undisturbed samples to determine the geotechnical properties of the soil.

**Table – 1: Items of the supplied samples**

| No | Name | Bore hole no | Depth | Quantities |
|---|---|---|---|---|
| 01 | Disturbed sample | 01 | 0-55 ft | 11 |
| 02 | Disturbed sample | 02 | 0-55 ft | 11 |
| 03 | Disturbed sample | 03 | 0-50 ft | 10 |
| 04 | Disturbed sample | - | - | - |
| 05 | Disturbed sample | - | - | - |
| 06 | Disturbed sample | - | - | - |
| 07 | Disturbed sample | - | - | - |
| 08 | Disturbed sample | - | - | - |
| 09 | Disturbed sample | - | - | - |
| 10 | Disturbed sample | - | - | - |

## Laboratory Classifications

The disturbed soil samples, supplied by the client were thoroughly examined and classified roughly (color and type) and noted in accordance with bore hole (BH) and depth (Please see enclosed Tables).

## Laboratory Works & Analysis

In addition to visual examination of the sample, Bearing Capacity of soil samples including skin friction & end bearing were calculated.

## Bearing capacity for shallow foundation

The N value gives useful information with regards to consistency of cohesive soil and relative density of cohesion less soil. These relationships are given in the following Tables for cohesive and cohesion less soil, respectively.

**Table 2: Relation of N-value with the Consistency and the Unconfined Compressive Strength of Cohesive Soils (after Bowles 1997)**

| N-values | Consistency | Unconfined Compressive Strength ($q_u$) | |
|---|---|---|---|
| | | tsf | kPa |
| 0-2 | Very soft | 0 to 0.25 | 0 to 25 |
| 3-5 | Soft | 0.25 to 0.50 | 25 to 50 |
| 6-9 | Medium | 0.50 to 1 | 50 to 100 |
| 10-16 | Stiff | 1 to 2 | 100 to 200 |
| 17-30 | Very stiff | 2 to 4 | 200 to 400 |
| >30 | Hard | >4 | >400 |

**Table 3: N and ϕ Relation to Relative Density for Fine Sand (after Bowles 1997)**

| N values | Compactness | Relative Density, Dr % | ϕ (degree) | Unit Weight (pcf) |
|---|---|---|---|---|
| 1 – 2 | Very Loose | 0 – 15 | 26 – 28 | 70 – 100 |
| 3 – 6 | Loose | 15 – 35 | 28 – 30 | 90 – 115 |
| 7 – 15 | Medium | 35 – 65 | 30 – 34 | 110 – 130 |
| 16 – 30 | Dense | 65 – 85 | 33 – 38 | 110 – 140 |
| >30 | Very Dense | >85 | >50 | 130 – 150 |

**Table 4: N and ϕ Relation to Relative Density for Medium Sand (after Bowles 1997)**

| N values | Compactness | Relative Density, Dr % | ϕ (degree) | Unit Weight (pcf) |
|---|---|---|---|---|
| 2 – 3 | Very Loose | 0 – 15 | 27 – 28 | 70 – 100 |
| 4 – 7 | Loose | 15 – 35 | 30 – 32 | 90 – 115 |
| 8 – 20 | Medium | 35 – 65 | 32 – 36 | 110 – 130 |
| 21 – 40 | Dense | 65 – 85 | 36 – 42 | 110 – 140 |

**Table 5: N and ϕ Relation to Relative Density for Coarse Sand (after Bowles 1997)**

| N values | Compactness | Relative Density, Dr % | ϕ (degree) | Unit Weight (pcf) |
|---|---|---|---|---|
| 3 – 6 | Very Loose | 0 – 15 | 28 – 30 | 70 – 100 |
| 5 – 9 | Loose | 15 – 35 | 30 – 34 | 90 – 115 |
| 10 – 25 | Medium | 35 – 65 | 33 – 40 | 110 – 130 |
| 26 – 40 | Dense | 65 – 85 | 40 – 50 | 110 – 140 |
| >40 | Very Dense | >85 | >50 | 130 – 150 |

**Bearing capacity for deep foundation**

Foundations, preferably piles are considered to transmit heavy load from superstructure to deep and hard strata. As the formation of soil strata consists of cohesive and non-cohesive type, the evaluation criteria for the determination of skin friction and the end bearing capacities of piles also vary. Details of the evaluation are provided in the following sub-headings.

**Table 6: Formula Used to Determine Bearing Capacity of Soil**

| **Shallow Foundation** | |
|---|---|
| For Cohesive Soil | |
| $q_a = q_u + \gamma \frac{D}{3}$<br>$q_a = q_u$(Common Practice) | Where, $q_a$=Allowable bearing capacity, $q_u$= Unconfined Compressive Strength, $\gamma$= Unit Weight of soil, $D$= Depth of footing |
| For Cohesion less Soil | |
| $q_{ult} = 30N;$<br>$q_a = q_{ult}/FS$ | $q_{ult}$= Ultimate bearing capacity in kPa, $N$ = SPT Number, $FS$= Factor of Safety |
| **Source: Bowles, J. E., "Physical and Geotechnical Properties of Soils", 2nd Edition** | |
| **Deep Foundation** | |
| For Cohesive Soil | |
| $f_{su} = \alpha C_u$ ;<br>$q_{pu} = N_c C_u$ | $\alpha$ = Adhesion factor, $f_{su}$ = Ultimate skin friction, $C_u$ = Undrained Shear Strength, $q_{pu}$ = Ultimate end bearing of pile, $N_c$ = Bearing capacity factor for deep foundation=9 |
| For Cohesion less Soil | |
| $f_{su} = N/50\ tsf$ ;<br>$q_{pu} = 4N\ tsf$ | $f_{su}$ = Ultimate skin friction, $q_{pu}$ = Ultimate end bearing of pile, $N$ = SPT Number |
| **Source: Rajapakse RA (2008), "Geotechnical Engineering Calculations and Rules of Thumb"** | |

Based on the above formula, the important parameters for foundation design have been determined and are given in **Table 7**, **Table 8, Table 9 and Table 10** respectively.

**Table 7: The allowable skin friction for Precast piles in Tsf (FS=2.5)**

| **Depth** | **Bore Hole** | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **ft** | **BH01** | **BH02** | **BH03** | **BH04** | **BH05** | **BH06** | **BH07** | **BH08** | **BH09** | **BH10** |
| **5** | 0.156 | 0.174 | 0.156 | - | - | - | - | - | - | - |
| **10** | 0.118 | 0.096 | 0.118 | - | - | - | - | - | - | - |
| **15** | 0.118 | 0.138 | 0.049 | - | - | - | - | - | - | - |
| **20** | 0.096 | 0.049 | 0.118 | - | - | - | - | - | - | - |
| **25** | 0.156 | 0.112 | 0.152 | - | - | - | - | - | - | - |
| **30** | 0.468 | 0.118 | 0.080 | - | - | - | - | - | - | - |
| **35** | 0.049 | 0.156 | 0.224 | - | - | - | - | - | - | - |
| **40** | 0.096 | 0.096 | 0.240 | - | - | - | - | - | - | - |
| **45** | 0.334 | 0.361 | 0.294 | - | - | - | - | - | - | - |
| **50** | 0.669 | 0.535 | 0.535 | - | - | - | - | - | - | - |
| **55** | 0.669 | 0.669 | - | - | - | - | - | - | - | - |
| **60** | - | - | - | - | - | - | - | - | - | - |
| **65** | - | - | - | - | - | - | - | - | - | - |
| **70** | - | - | - | - | - | - | - | - | - | - |
| **75** | - | - | - | - | - | - | - | - | - | - |
| **80** | - | - | - | - | - | - | - | - | - | - |
| **85** | - | - | - | - | - | - | - | - | - | - |
| **90** | - | - | - | - | - | - | - | - | - | - |
| **95** | - | - | - | - | - | - | - | - | - | - |
| **100** | - | - | - | - | - | - | - | - | - | - |

**Table 8: The allowable end bearing for Precast piles in Tsf (FS=2.5)**

| Depth | Bore Hole | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| ft | BH01 | BH02 | BH03 | BH04 | BH05 | BH06 | BH07 | BH08 | BH09 | BH10 |
| 5 | 1.575 | 1.800 | 1.575 | - | - | - | - | - | - | - |
| 10 | 1.125 | 0.900 | 1.125 | - | - | - | - | - | - | - |
| 15 | 1.125 | 1.350 | 0.450 | - | - | - | - | - | - | - |
| 20 | 0.900 | 0.450 | 1.125 | - | - | - | - | - | - | - |
| 25 | 1.575 | 22.400 | 30.400 | - | - | - | - | - | - | - |
| 30 | 7.875 | 1.125 | 16.000 | - | - | - | - | - | - | - |
| 35 | 0.450 | 1.575 | 44.800 | - | - | - | - | - | - | - |
| 40 | 0.900 | 0.900 | 3.825 | - | - | - | - | - | - | - |
| 45 | 5.625 | 6.075 | 4.950 | - | - | - | - | - | - | - |
| 50 | 11.250 | 9.000 | 9.000 | - | - | - | - | - | - | - |
| 55 | 11.250 | 11.250 | - | - | - | - | - | - | - | - |
| 60 | - | - | - | - | - | - | - | - | - | - |
| 65 | - | - | - | - | - | - | - | - | - | - |
| 70 | - | - | - | - | - | - | - | - | - | - |
| 75 | - | - | - | - | - | - | - | - | - | - |
| 80 | - | - | - | - | - | - | - | - | - | - |
| 85 | - | - | - | - | - | - | - | - | - | - |
| 90 | - | - | - | - | - | - | - | - | - | - |
| 95 | - | - | - | - | - | - | - | - | - | - |
| 100 | - | - | - | - | - | - | - | - | - | - |

**Table 9: The allowable skin friction for Bored piles in Tsf (FS=2.5)**

| Depth | Bore Hole | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| ft | BH01 | BH02 | BH03 | BH04 | BH05 | BH06 | BH07 | BH08 | BH09 | BH10 |
| 5 | 0.101 | 0.113 | 0.101 | - | - | - | - | - | - | - |
| 10 | 0.077 | 0.062 | 0.077 | - | - | - | - | - | - | - |
| 15 | 0.077 | 0.090 | 0.032 | - | - | - | - | - | - | - |
| 20 | 0.062 | 0.032 | 0.077 | - | - | - | - | - | - | - |
| 25 | 0.101 | 0.073 | 0.099 | - | - | - | - | - | - | - |
| 30 | 0.304 | 0.077 | 0.052 | - | - | - | - | - | - | - |
| 35 | 0.032 | 0.101 | 0.146 | - | - | - | - | - | - | - |
| 40 | 0.062 | 0.062 | 0.156 | - | - | - | - | - | - | - |
| 45 | 0.217 | 0.235 | 0.191 | - | - | - | - | - | - | - |
| 50 | 0.435 | 0.348 | 0.348 | - | - | - | - | - | - | - |
| 55 | 0.435 | 0.435 | - | - | - | - | - | - | - | - |
| 60 | - | - | - | - | - | - | - | - | - | - |
| 65 | - | - | - | - | - | - | - | - | - | - |
| 70 | - | - | - | - | - | - | - | - | - | - |
| 75 | - | - | - | - | - | - | - | - | - | - |
| 80 | - | - | - | - | - | - | - | - | - | - |
| 85 | - | - | - | - | - | - | - | - | - | - |
| 90 | - | - | - | - | - | - | - | - | - | - |
| 95 | - | - | - | - | - | - | - | - | - | - |
| 100 | - | - | - | - | - | - | - | - | - | - |

**Table 10: The allowable end bearing for Bored piles in Tsf (FS=2.5)**

| Depth | Bore Hole | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| ft | BH01 | BH02 | BH03 | BH04 | BH05 | BH06 | BH07 | BH08 | BH09 | BH10 |
| 5 | 0.709 | 0.810 | 0.709 | - | - | - | - | - | - | - |
| 10 | 0.506 | 0.405 | 0.506 | - | - | - | - | - | - | - |
| 15 | 0.506 | 0.608 | 0.203 | - | - | - | - | - | - | - |
| 20 | 0.405 | 0.203 | 0.506 | - | - | - | - | - | - | - |
| 25 | 0.709 | 10.080 | 13.680 | - | - | - | - | - | - | - |
| 30 | 3.544 | 0.506 | 7.200 | - | - | - | - | - | - | - |
| 35 | 0.203 | 0.709 | 20.160 | - | - | - | - | - | - | - |
| 40 | 0.405 | 0.405 | 1.721 | - | - | - | - | - | - | - |
| 45 | 2.531 | 2.734 | 2.228 | - | - | - | - | - | - | - |
| 50 | 5.063 | 4.050 | 4.050 | - | - | - | - | - | - | - |
| 55 | 5.063 | 5.063 | - | - | - | - | - | - | - | - |
| 60 | - | - | - | - | - | - | - | - | - | - |
| 65 | - | - | - | - | - | - | - | - | - | - |
| 70 | - | - | - | - | - | - | - | - | - | - |
| 75 | - | - | - | - | - | - | - | - | - | - |
| 80 | - | - | - | - | - | - | - | - | - | - |
| 85 | - | - | - | - | - | - | - | - | - | - |
| 90 | - | - | - | - | - | - | - | - | - | - |
| 95 | - | - | - | - | - | - | - | - | - | - |
| 100 | - | - | - | - | - | - | - | - | - | - |

## CONCLUSION AND RECOMMENDATION

On the basis of analysis and information provided by the owner following recommendations are suggested for the proposed five storied residential building.

***Soil profile scenario***

Overall soil information of the investigated site is irregular in between the boreholes and vertical inconsistency exists in the soil profile.

Clay soil layer (brown/grey) with inadequate bearing capacity has been identified at the top at boreholes 1, 2 and 3. In the case of borehole 1, a grey stiff clay layer with moderate bearing capacity has been observed at 30 ft depth from EGL, however, a layer of grey soft clay with low bearing capacity has been found between 35-45 ft depth, which in turn making the investigation site vulnerable. In the case of borehole 2, grey fine sand (with silt and clay) layer with moderate bearing capacity has been observed at 25 ft depth, however, a layer of grey soft clay of low bearing capacity has been found at 40 ft depth. In the case of borehole 3, grey fine sand (with silt and clay) layer with moderate bearing capacity has been found at 25ft depth, however, a fall of the soil bearing capacity has been observed at 30 ft depth from EGL. Though grey hard/stiff clay with moderate to good bearing capacity has been found from 45 ft depth from EGL regardless of borehole location, the scenario is unknown beyond 55 ft depth (50 ft for borehole 3), as the client did not provide any data beyond the above-mentioned depth.

From the above discussion, it is very clear that the whole scenario of the proposed site is variable in nature.

***Foundation options***

The deep foundation (bored/precast pile) may be selected to ensure the stability of the structure and to overcome the ground improvement complication. For designing any suitable deep foundation, the necessary data for skin friction and end bearing can be found from the attached table in the report (**table 7-10**) which is based on existing soil conditions.

***Safety provisions***

Since the proposed site is composed of alternate layers of soft cohesive and granular soil, it is reasonably vulnerable to differential settlement. Therefore, it is strongly recommended to check the **settlement** of the site considering building load while designing deep foundation. Obviously **pile load test** covering the entire project area must be carried out for confirming the design parameters of the service pile. Moreover, the earthquake vulnerability of the project area as well as the quality management of the construction work including workmanship should be taken into account for the safe design of shallow or deep foundation.

***Attachments***

The layout of the boreholes of the investigated site and the bore-log profiles with SPT values are shown in the Appendix.

***Special notes***

These suggestions are absolutely tentative and selection of type, depth and dimensions of foundation would be decided by the Consultant or Design Engineer. It is also to be noted that the above recommendation is restricted only to the existing soil condition of **50-55** ft as client provided.

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